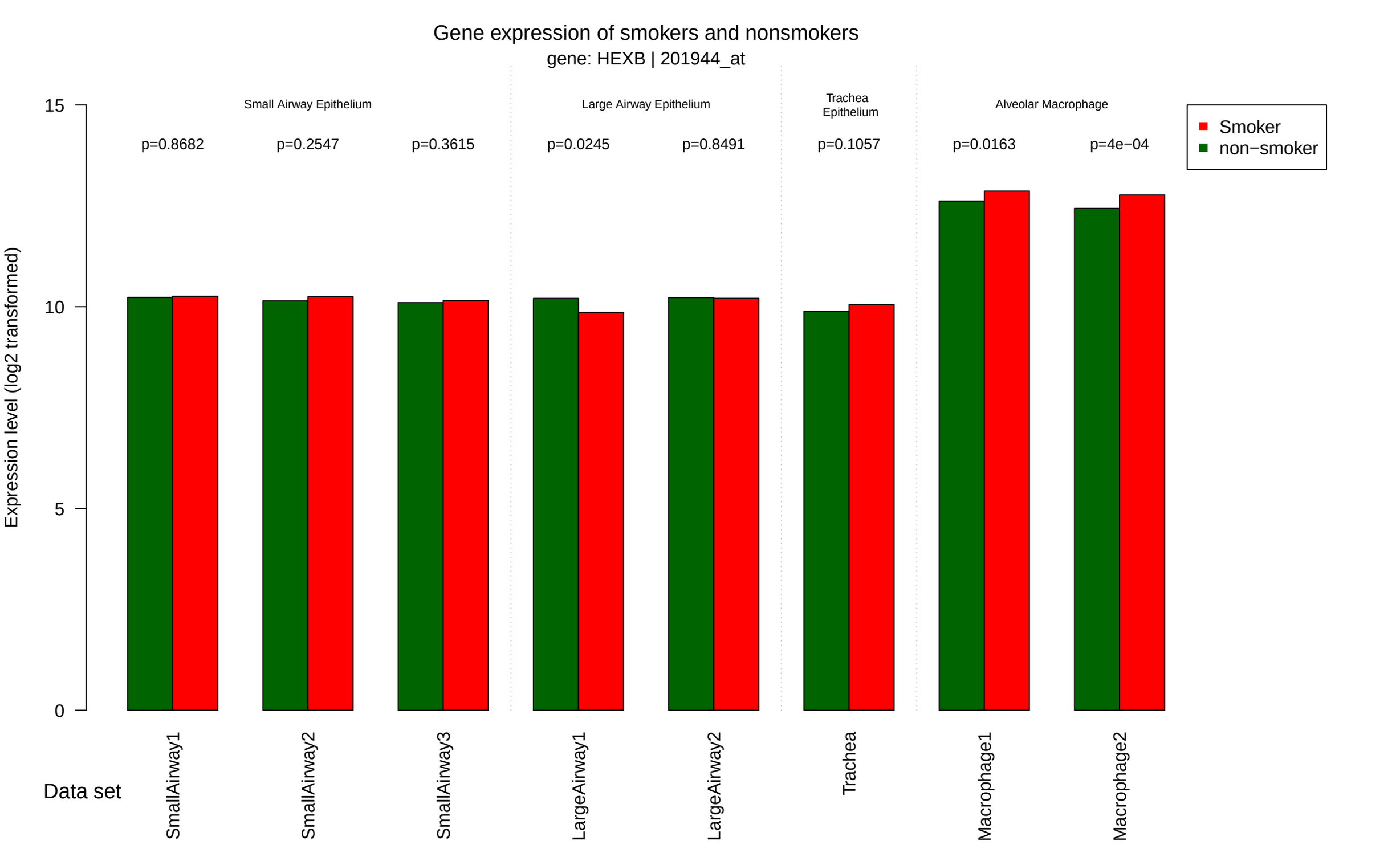
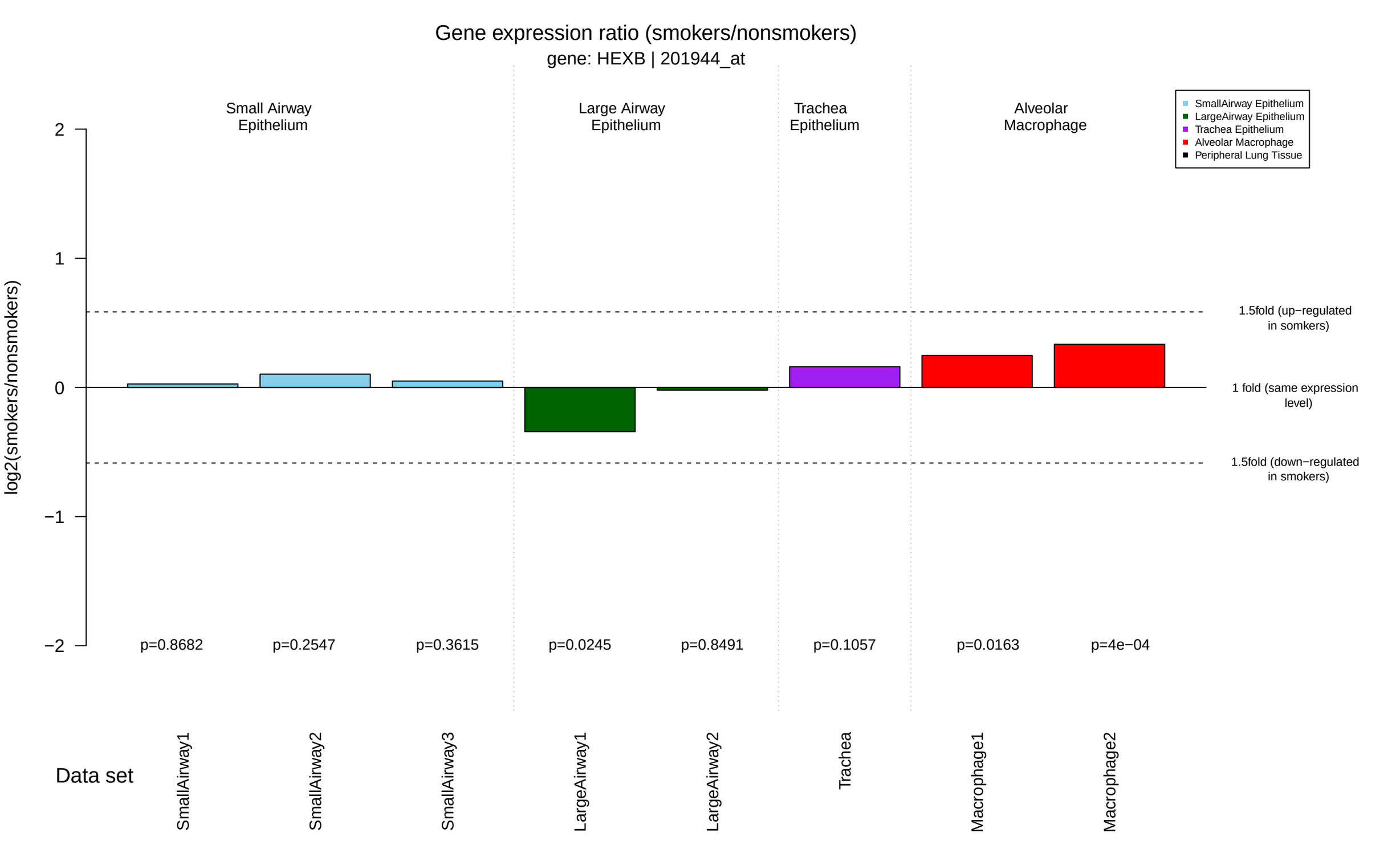
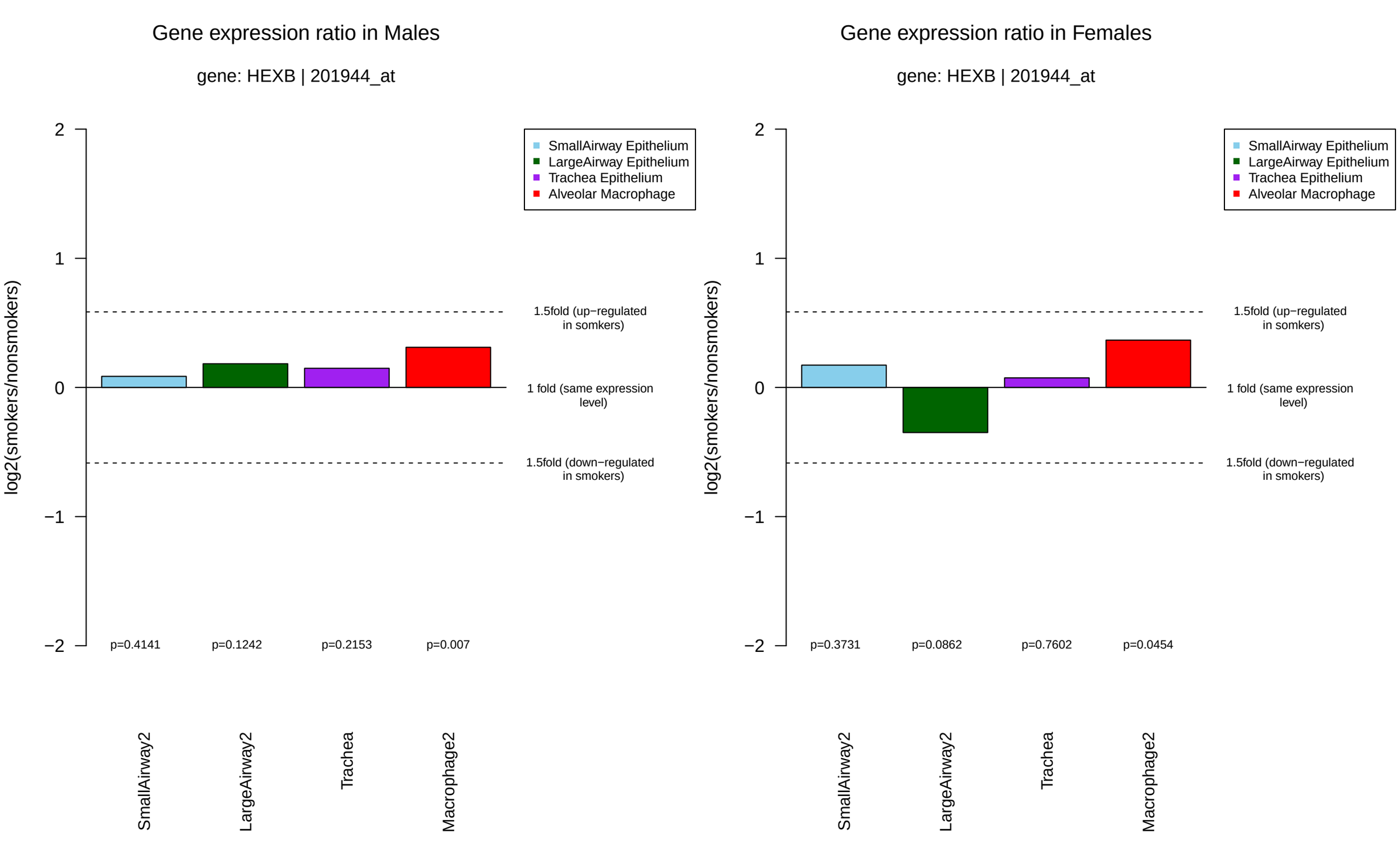
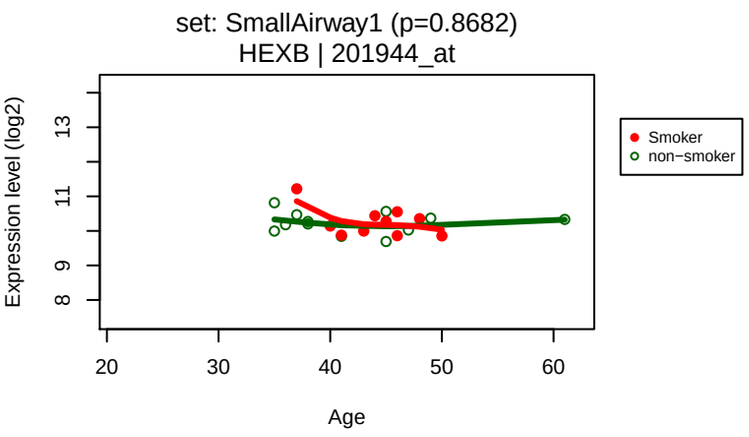
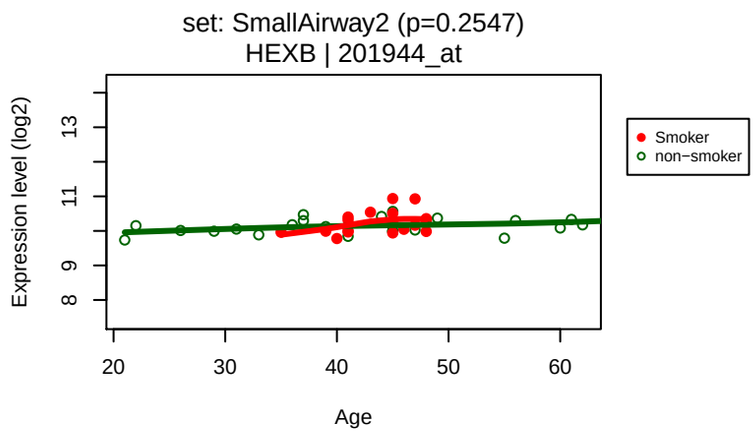
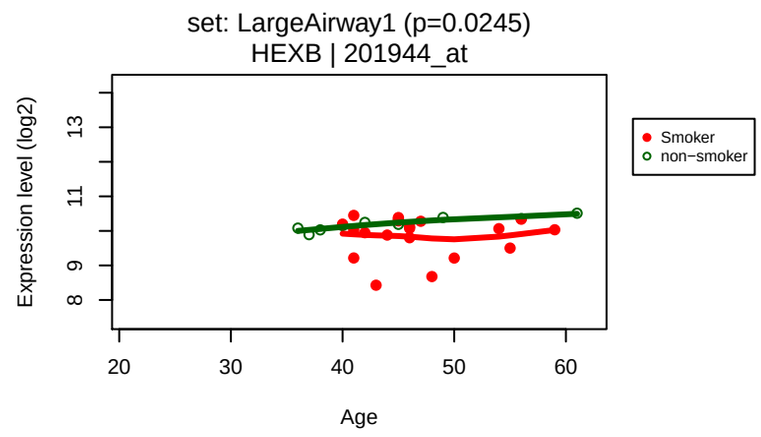

Gene expression of smokers and nonsmokers
gene: HEXB | 201944_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.8682
p=0.2547
p=0.3615
p=0.0245
p=0.8491
p=0.1057
p=0.0163
p=4e−04
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

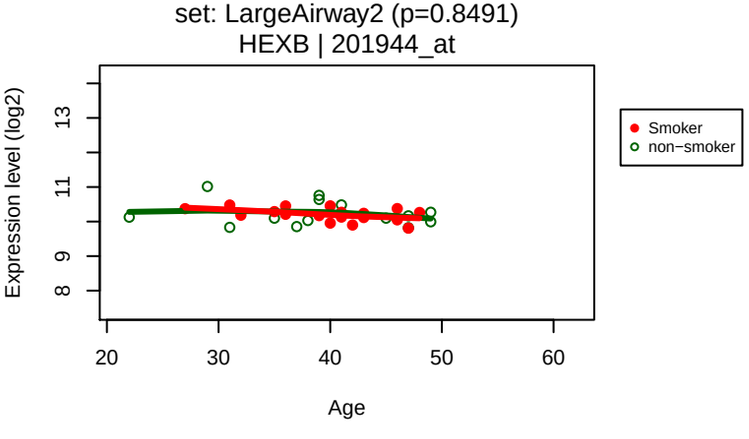

Gene expression ratio (smokers/nonsmokers)
gene: HEXB | 201944_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8682
p=0.2547
p=0.3615
p=0.0245
p=0.8491
p=0.1057
p=0.0163
p=4e−04
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

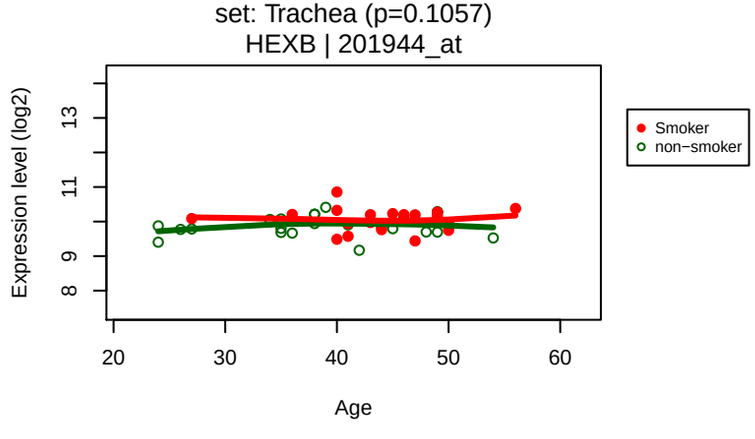

Gene expression ratio in Males
gene: HEXB | 201944_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4141
p=0.1242
p=0.2153
p=0.007
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HEXB | 201944_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3731
p=0.0862
p=0.7602
p=0.0454
SmallAirway2
LargeAirway2
Trachea
Macrophage2

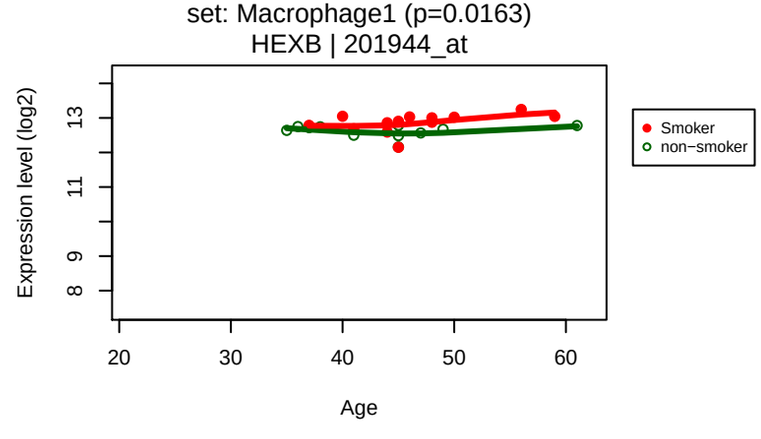

set: SmallAirway1 (p=0.8682)
HEXB | 201944_at
Expression level (log2)
8
9
11
13
20
30
40
50
60
Age
Smoker
non−smoker

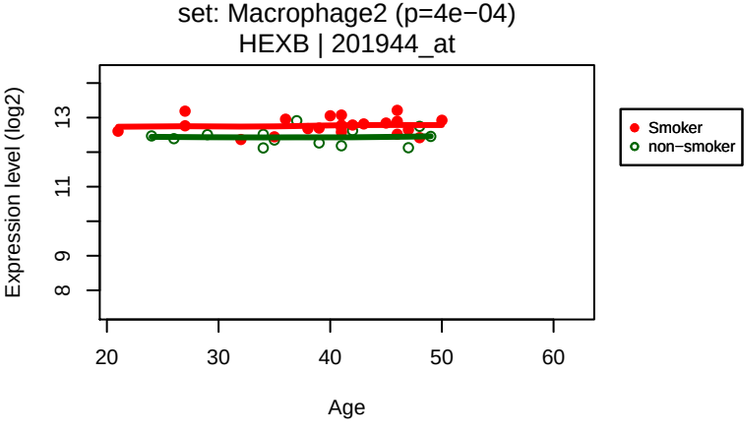

set: SmallAirway2 (p=0.2547)
HEXB | 201944_at
Expression level (log2)
8
9
11
13
20
30
40
50
60
Age
Smoker
non−smoker

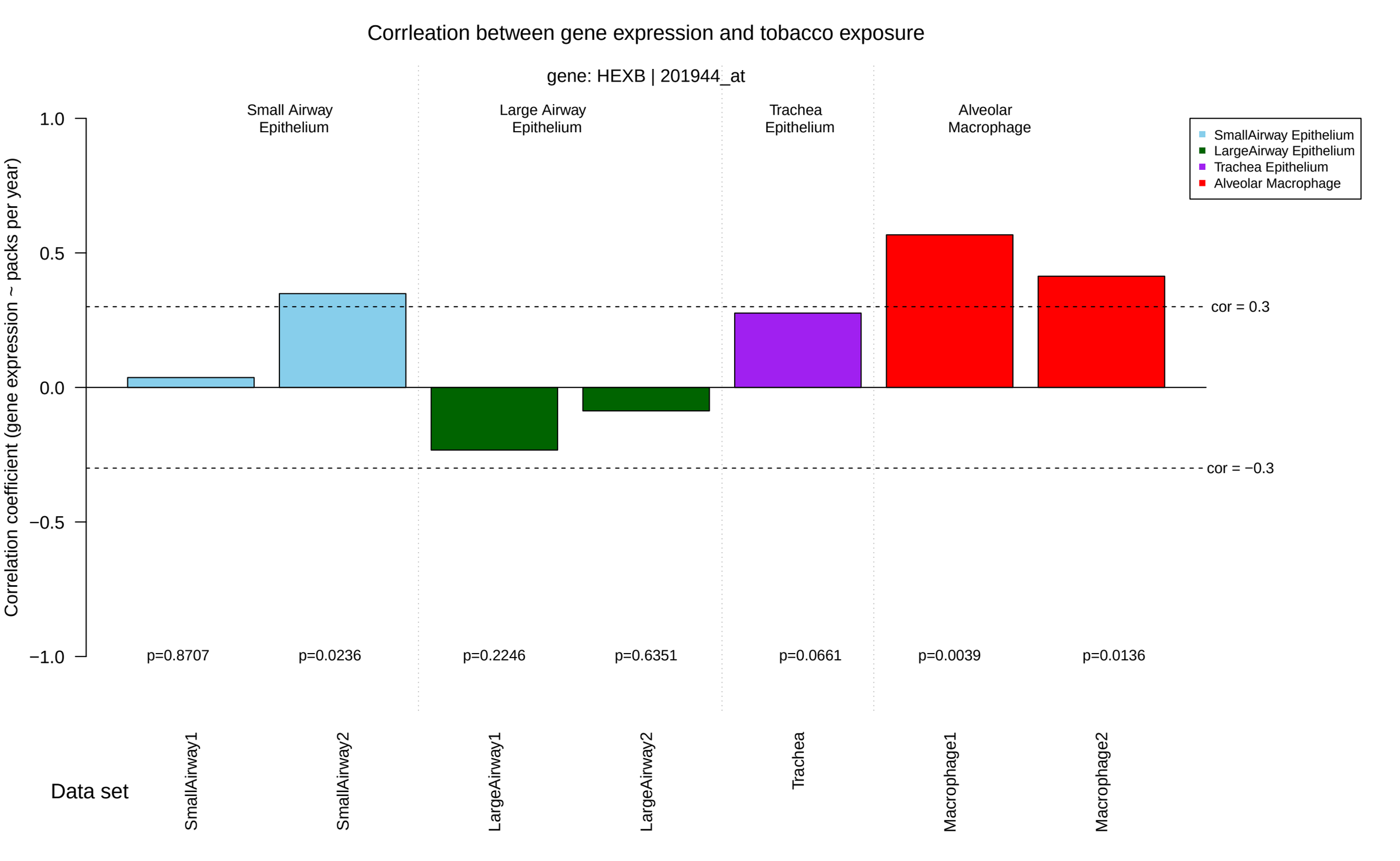

set: LargeAirway1 (p=0.0245)
HEXB | 201944_at
Expression level (log2)
8
9
11
13
20
30
40
50
60
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8491)
HEXB | 201944_at
Expression level (log2)
8
9
11
13
20
30
40
50
60
Age
Smoker
non−smoker


set: Trachea (p=0.1057)
HEXB | 201944_at
Expression level (log2)
8
9
11
13
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0163)
HEXB | 201944_at
Expression level (log2)
8
9
11
13
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage2 (p=4e−04)
HEXB | 201944_at
Expression level (log2)
8
9
11
13
20
30
40
50
60
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HEXB | 201944_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8707
p=0.0236
p=0.2246
p=0.6351
p=0.0661
p=0.0039
p=0.0136
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2